Data-Driven Modeling of Solar Photovoltaic Degradation

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This report analyzes solar photovoltaic (PV) performance degradation over time from 2011 to 2023, corrected for temperature and irradiance. The regression model results are compared to published findings for the same system from 1994–2003. The degradation follows a linear downward trend around -1.5% per year, which is 50% faster than the literature value, potentially due to compounding degradation over the years. The weather correction factor is also tested to determine its effectiveness at eliminating the influence of temperature and irradiance fluctuations. Their influence is found to be small yet statistically significant.

I. Nomenclature

PR_{corr}	=	weather-corrected performance ratio
$EN_{AC,i}$	=	measured AC generation from the solar power system (W)
P_{STC}	=	total system power rating (W)
G_{POA}	=	plane-of-array irradiance at the solar array (W/m^2)
G_{STC}	=	standard test (STC) plane-of-array irradiance (1000 W/m^2)
i	=	point in time (for this project, the units are days)
$T_{cell,i}$	=	average cell temperature for one day (°C)
$T_{cell-typ-avg}$	=	average cell temperature for a year (°C)
δ	=	power-temperature coefficient for the solar modules (%/°C, negative)

II. Introduction

SOLAR photovoltaic (PV) power systems play a key role in the global transition away from non-renewable fossil fuel energy sources. In 2025, US solar PV supplied 297.4 gigawatt-hours of energy, a 137% increase over the 21.7 gigawatt-hours produced in 2015 [1]. In 2026, solar provided 11.6% of total US electricity [2]. Solar power systems have unique advantages including low cost, zero moving parts, no noise pollution, and zero operational greenhouse gas emissions. These benefits come with an important caveat: solar cells experience a degree of irreversible degradation over time, which may require upgrades. Some factors, such as soiling, produce *recoverable* degradation losses, while others are intrinsic to the panel, and may lead to the system requiring replacement. Thus, it is important for consumers and solar companies to understand how the performance of the array is declining so that appropriate maintenance schedules and replacement timelines can be determined and costs can be properly estimated.

This report covers a data-driven process for measuring performance degradation using information from the SERF-EAST system at the National Laboratory of the Rockies (NRL) in Golden Colorado. Data are extracted from the PVDAQ database [3], and the system is analyzed using least squares and robust linear regression to model the performance ratio (PR) with respect to temperature, irradiance, and time. The analysis uses hourly data collected over 12 years between 2011 and 2023. Results are compared with values obtained in the literature for the same system between 1994 and 2002 [4].

III. Background

Solar PV systems consist of solar cells connected in an array that convert energy from incident photons into electrical current. In terrestrial applications, the current is passed through an inverter to convert it to alternating current (AC). The conversion efficiency of solar cells from incident radiation to electrical energy is with about 15–22% with a theoretical maximum of 30% [5], and the operating efficiency of the inverter is 90–95% [6]. Since PV systems are exposed to harsh weather and day/night cycles, their output is intermittent and subject to environmental damage and soiling. Losses include heat dissipated by the solar array (thermalization loss), heat dissipated by the inverter, line losses, cell mismatch, sunlight reflected from the arrays, and failure of components [7]. These losses result in a lower AC power output than

the value reported by the solar panel manufacturer. Underlying the primary loss components is the intrinsic degradation of the solar cells. The ratio of actual measured AC power to nameplate AC power is the performance ratio (PR), which is used to quantify the overall effectiveness of the system. Tracking the PR over time can provide a useful indicator of system degradation.

IV. Technical Approach

The first step in degradation analysis of the SERF-EAST system was to define the performance ratio (PR), which compares the actual power generation to a standardized rating. In this project, the PR is calculated using the weather-corrected formula from ref. [7], as shown in Equation 1.

$$PR_{corr} = \frac{\sum_i EN_{AC,i}}{\sum_i \left[P_{STC} \left(\frac{G_{POA}}{G_{STC}} \right) (0.01\delta(T_{cell-typ-avg-T_{cell,i}}) \right]} \tag{1}$$

The corrected PR is designed to eliminate the influence of cell temperature and irradiance on the measured power output, assuming linear relationships between parameters and power output. The outdoor power rating (P_{STC}) of 6140 W AC at standard temperature 25°C and irradiance 1000 W/m^2 was determined from ref. [8]. The typical average cell temperature $T_{cell-typ-avg}$ of 32.65 °C was found by taking the average of daily mean temperatures from 2011 to 2022. The temperature coefficient δ may vary, however, for this project, it is assumed to be 0.5%/°C based on the typical value stated in ref. [7]. The dataset for SERF-EAST includes cell temperatures, AC and DC power outputs, and plane-of-array (POA) irradiance from 1994 to 2023 with gaps in 2006 and 2010. Missing data are attributed to inverter outages, sensor failures, and maintenance [4]. While AC power, DC power, and cell temperatures are consistently populated, the PVDAQ recorded POA irradiance data are sparse. Hence, satellite irradiance data from the National Solar Radiation Database (NSRDB) at the known latitude, longitude, tilt angle, and azimuth for SERF-EAST are applied instead. Hourly average values for the desired date range are obtained using the NSRDB Physical Solar Model (PSM) v4 aggregated values [9]. This dataset is derived from a Geostationary Operational Environmental Satellite (GOES), which recorded images of the Earth's atmosphere from space. For this project, the satellite image data are translated into POA irradiance time-history via PVlib using the Perez Sky-Diffuse model, which accounts for the position of the sun, reflection from the Earth's surface, and scattering of light through the atmosphere [10].

The next step is to reduce the datasets to manageable size and quality for plotting. The PVDAQ data is filtered to remove values corresponding to AC power generation less than or equal to zero. The dataset is then reduced from minute-by-minute daily values to monthly sums of power generation and corrected STC power generation along with average cell temperature and POA irradiance. The monthly data are then used to compute PR.

Initially, samples were used from 2001 to 2023, however the PR was found to decrease until 2006, after which it increased (see Figure 1). Data from 2006 and 2007 were sparse, thus it was elected to sample only from 2011 onward to remove anomalies, likely due to maintenance and upgrades undertaken during missing or sparse years.

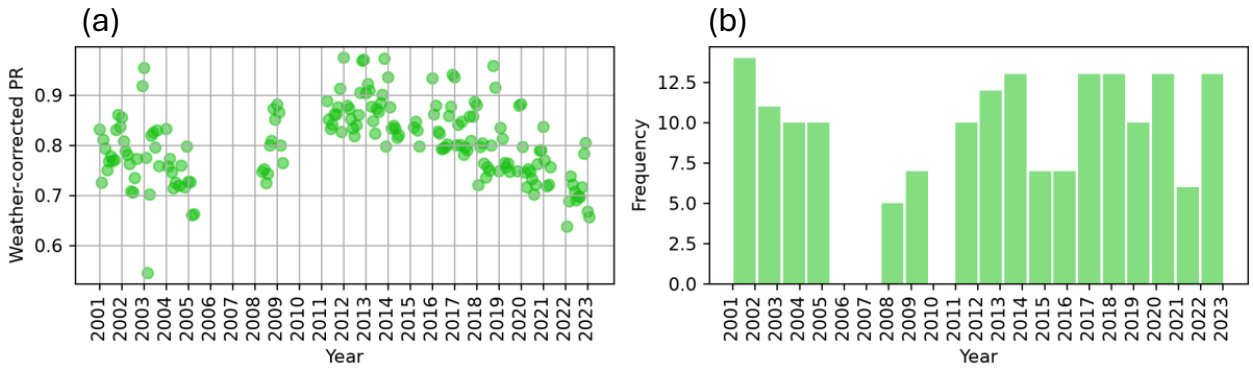


Fig. 1 Plot shows non linear variation in PR between 2008 and 2011 (a) Plot of corrected PR over time from 2001 to 2023. (b) Frequency refers to the number of months of available data with positive AC generation in a given year

The next step is to build the regression model from the filtered data. Importantly, outlier values of PR are still present. Two methods are tested to minimize the influence of these outliers on the ordinary least-squares (OLS) regression. The first method is to only use parameter (x) values and PR (y) values within the interquartile range between quartiles Q1 (0.15) and Q3 (0.85), assuming normally-distributed data. This approach risks losing significant points near the minimum and maximum PR. A better solution is to apply a robust fitting algorithm in place of OLS. The Robust Linear Model (RLM) algorithm included in SciPy Statsmodels uses a technique called M-estimation to reduce the effect of outliers [11]. In M-estimation, the regression is derived by minimizing the sum of a symmetric function $\rho(r_i)$ of the residual. This project uses Huber's M-estimator, as described in [12]. This method iteratively determines the optimal coefficients by adjusting terms in the ρ function until convergence.

V. Results

The two regression models (OLS and RLM) are fit to the PR data as a function of average monthly cell temperature and POA irradiance, with and without IQR-based filtering. A 3-D hyperplane, $PR = \beta_T T + \beta_I I + \beta_t t$ is produced, where β values are the model coefficients, T is the average monthly cell temperature, I is the average monthly POA irradiance, and t is the time since 2011 in months. Plots showing the behavior without filtering outliers are shown in Figure 2(a), (c), and (e). Corresponding histograms in Figure 2(b), (d), and (f) are included to show the relative numbers of samples within each temperature and irradiance block and year. The models include a total of 120 observations each.

First, the trend in PR versus time is assessed, assuming a threshold p-value of 0.05 below which the null hypothesis $H_0 : \beta_t = 0$ can be rejected. The model is a slice $PR = \beta_T \bar{T} + \beta_I \bar{I} + \beta_t t$ of the overall model hyperplane. The p-value for the slope in the OLS model, as determined by a t-test, is 0.001, indicating that the trend is statistically significant. For the RLM fit, the p-value is <0.0001, which also indicates a strong correlation. The slopes, shown in Figure 2(e), give the approximate yearly degradation rate between 2011 and 2023 of -2.05%/year from OLS and -1.50%/year from RLM. These values have greater magnitudes than the -1%/year degradation rate reported for 1994 to 2002 in the literature [4]. Greater values of degradation rate from 2011 to 2023 indicate damage accumulation over time. Since outliers are still present in the data, the overall R-squared value is 0.114 for OLS without IQR-based filtering.

Analysis of variance and individual coefficient hypothesis tests are used to determine how effectively the weather-corrected PR normalization eliminates the influence of variation in irradiance and temperature. The OLS and RLM curves, $PR = \beta_T T + \beta_I \bar{I} + \beta_t \bar{t}$ and $PR = \beta_T \bar{T} + \beta_I I + \beta_t \bar{t}$, are plotted over measured PR versus temperature and irradiance, respectively in Figure 2(a) and (c). The OLS trend in PR versus average cell temperature is positive due to the influence of outliers. On its own, this suggests, incorrectly, that performance increases at higher temperatures. From OLS, the p-values for the temperature and irradiance curves are 0.432 and 0.049, respectively. For a probability threshold of 0.05, these results indicate no significance in the trend for temperature and almost no significance in the trend for irradiance. When the influence of outliers is minimized via RLM, the resulting p-values (from a z-score test) are 0.039 and 0.013 for cell temperature and POA irradiance, respectively, suggesting a statistically significant trend in each. Analysis of variance is performed to compare OLS models with and without temperature and irradiance terms (both are removed simultaneously in the reduced model). The resulting F-statistic value is 4.705 and the corresponding p-value is 0.011, indicating a significant influence of the two parameters. The trends in PR as a function of temperature and irradiance are -0.17% per °C and 0.02% per W/m^2, respectively. The temperature trend may be a result of incorrect specification of the thermal constant in Equation 1, while the irradiance trend is potentially the consequence of underlying nonlinearity in the power output as a function of irradiance. It is demonstrated in ref. [7] that the weather-corrected PR is less prone to seasonal variation than a temperature-independent PR, however the results of the present report show that not all thermal and irradiance effects are removed by the normalization.

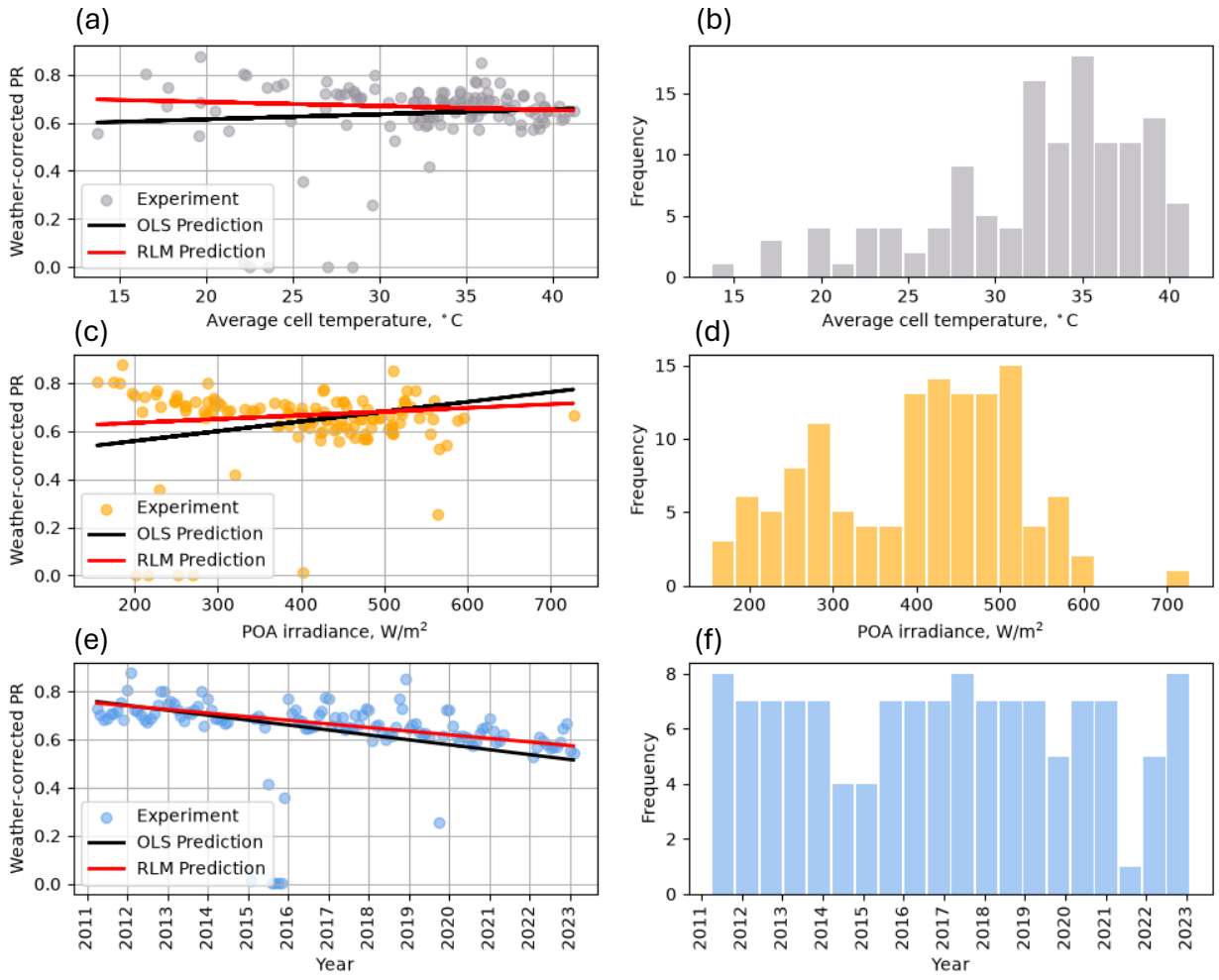


Fig. 2 Plots and histograms for each slice of the multiple linear regression. (a) and (b) show OLS and RLM predictions of the weather-corrected PR as a function of average cell temperature with frequency representing the number of data points with a given temperature. Similarly (c) and (d) are the plot and histogram for weather-corrected PR as a function of POA solar irradiance. Finally, (e) and (f) show the degradation trend over time and histogram representing the number of months containing valid data in each year.

The analysis is repeated with PR data points outside the 0.15-0.85 IQR taken as outliers and removed. The resulting plots and histograms are shown in Figure 3. After filtering, the OLS R-squared value is increased to 0.426, and the OLS line more closely matches the RLM line in each plotted slice. The degradation rates predicted by OLS and RLM, respectively, are -1.52%/year and -1.42%/year, lower than the previously predicted values of -2.05%/year and -1.50%/year. The change in the RLM value indicates that the outliers still had an influence on the robust model. The influence of average cell temperature and irradiance is also affected by the IQR-based filtering. The same analysis of variance is performed as before, yielding an F-statistic of 3.299 and p-value of 0.040, indicating less influence of temperature and irradiance than predicted by the unfiltered test. The coefficient p-values for temperature, irradiance, and time are 0.014, 0.063, and <0.0001, respectively for the OLS model, which suggests that all parameters are statistically significant except the POA irradiance. Similarly, the z-test for RLM produces p-values less than 0.0001 for all coefficients except irradiance, which has a p-value of 0.025. Thus, even with outlier filtering, the model shows temperature and

irradiance are statistically influential, despite normalization. The rates of change of PR with respect to temperature and irradiance (based on RLM results) are -0.29% per °C and 0.0001% per W/m^2, respectively. In the case of irradiance, this trend is negligible, even if it is statistically significant.

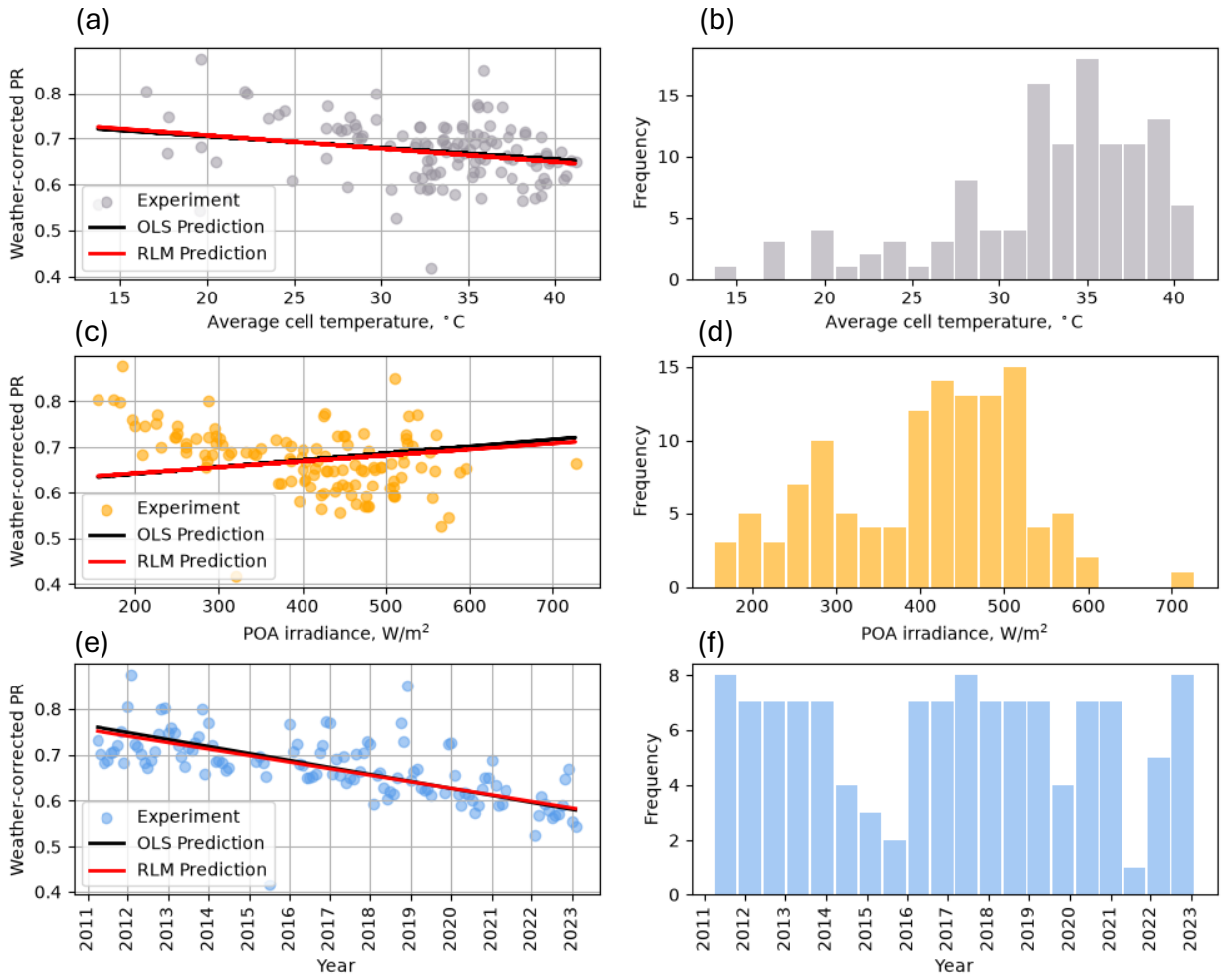


Fig. 3 Plots and histograms for each slice of the multiple linear regression with IQR-based outlier filtering. (a) and (b) show OLS and RLM predictions of the weather-corrected PR as a function of average cell temperature with frequency representing the number of data points with a given temperature. Similarly (c) and (d) are the plot and histogram for weather-corrected PR as a function of POA solar irradiance. Finally, (e) and (f) show the degradation trend over time and histogram representing the number of months containing valid data in each year.

VI. Conclusion

Data collected from 2011–2023 on the solar power system "SERF-EAST" at the National Laboratory of the Rockies in Golden, Colorado, along with satellite-based irradiance information from the National Solar Radiation Database were used to evaluate the degradation rate and test the effectiveness of temperature and plane-of-array irradiance normalization at correcting for seasonal variations. The results showed statistically significant but small trends (-0.29%/°C and in

the normalized performance metric (performance ratio) as a function of temperature and irradiance. Degradation rates were found to be 40-50% higher than in the literature, likely due to the available published research applying data from 1994–2002, when the newly-installed panels would have shown better performance and fewer compounded degradation effects. The current work expanded on prior research, showing that the normalized performance ratio effectively corrects for thermal and irradiance effects, and that 9 years after original data reporting, the system showed accelerated degradation.

References

[1] U.S. Energy Information Administration, "Electric Power Annual: Table 3.1.B. Net Generation from Renewable Sources: Total (All Sectors), 2015–2025," https://www.eia.gov/electricity/annual/table.php?t=epa_03_01_b.html, 2026. Accessed October 29, 2026.

[2] U.S. Energy Information Administration, "Electric Power Monthly: Table 1.1.B. Net Generation by State by Type of Producer by Energy Source," https://www.eia.gov/electricity/monthly/epm_table_grapher.php?t=epmt_es1b, 2026. Accessed October 29, 2026.

[3] Deline, C., Perry, K., Deceglie, M., Muller, M., Sekulic, W., and Jordan, D., "Photovoltaic Data Acquisition (PVDAQ) Public Datasets," Open Energy Data Initiative (OEDI), National Renewable Energy Laboratory, 2021. https://doi.org/10.25984/1846021, URL https://data.openei.org/submissions/4568, accessed October 29, 2026.

[4] Marion, B., and Adelstein, J., "Long-Term Performance of the SERF PV Systems," Tech. Rep. NREL/CP-520-33531, National Renewable Energy Laboratory, Golden, CO, May 2003.

[5] Zanatta, A. R., "The Shockley–Queisser limit and the conversion efficiency of silicon-based solar cells," *Results in Optics*, Vol. 9, 2022, p. 100320. https://doi.org/10.1016/j.rio.2022.100320, URL https://www.sciencedirect.com/science/article/pii/S2666950122001092.

[6] Fedkin, M., "6.5. Efficiency of Inverters," https://courses.ems.psu.edu/eme812/node/738, 2026. EME 812: Utility Solar Electric and Concentration, The Pennsylvania State University, accessed November 1, 2026.

[7] Dierauf, T., Growitz, A., Kurtz, S., Becerra Cruz, J. L., Riley, E., and Hansen, C., "Weather-Corrected Performance Ratio," Tech. Rep. NREL/TP-5200-57991, National Renewable Energy Laboratory, Golden, CO, Apr. 2013.

[8] Dyk, E. E., Strand, T., and Hansen, R., "SERF Photovoltaic Systems: Technical Report on System Performance for the Period, August 1, 1994–July 31, 1995," Tech. Rep. NREL/TP-411-21285, National Renewable Energy Laboratory, Golden, CO, Jun. 1996. https://doi.org/10.2172/258185, URL https://doi.org/10.2172/258185.

[9] National Laboratory of the Rockies, "National Solar Radiation Database: U.S. Data," https://nsrdb.nlr.gov/data-sets/us-data, 2016. Accessed October 31, 2026.

[10] Perez, R., Ineichen, P., Moore, K., Kmiecik, M., Chain, C., George, R., and Vignola, F., "A New Operational Satellite-to-Irradiance Model," *Solar Energy*, Vol. 73, No. 5, 2002, pp. 307–317. https://doi.org/10.1016/S0038-092X(02)00122-6.

[11] Statsmodels Developers, "M-Estimators for Robust Linear Modeling," https://www.statsmodels.org/stable/examples/notebooks/generated/robust_models_1.html, 2025. Statsmodels documentation, accessed October 31, 2026.

[12] Huber, P. J., *Robust Statistics*, John Wiley & Sons, 1981.